

WASP-104b

R-2665

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150207 · created 2026-08-30T00:44:59+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457060.87858 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.043
Transit depth (Rp/Rs)^2	2.24 +/- 1.29 [%]
Orbital Inclination (inc)	84.39 +/- 2.86
Airmass coefficient 1 (a1)	1.04 +/- 0.61
Airmass coefficient 2 (a2)	-0.05 +/- 1.53
Scatter in the residuals of the lightcurve fit is	57.62 %
Best Comparison Star	#3 - [622, 38]
Optimal Aperture	3.83
Optimal Annulus	6.69
Transit Duration (day)	0.075 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.043 vs 0.1214 (deviation 0.67σ)
Duration fit vs published	0.075 d vs 0.0742 d (deviation 0.03σ)
Mid-time O–C	0.6 min
Depth SNR	3.5
Residual scatter	57.62 %

Light curve

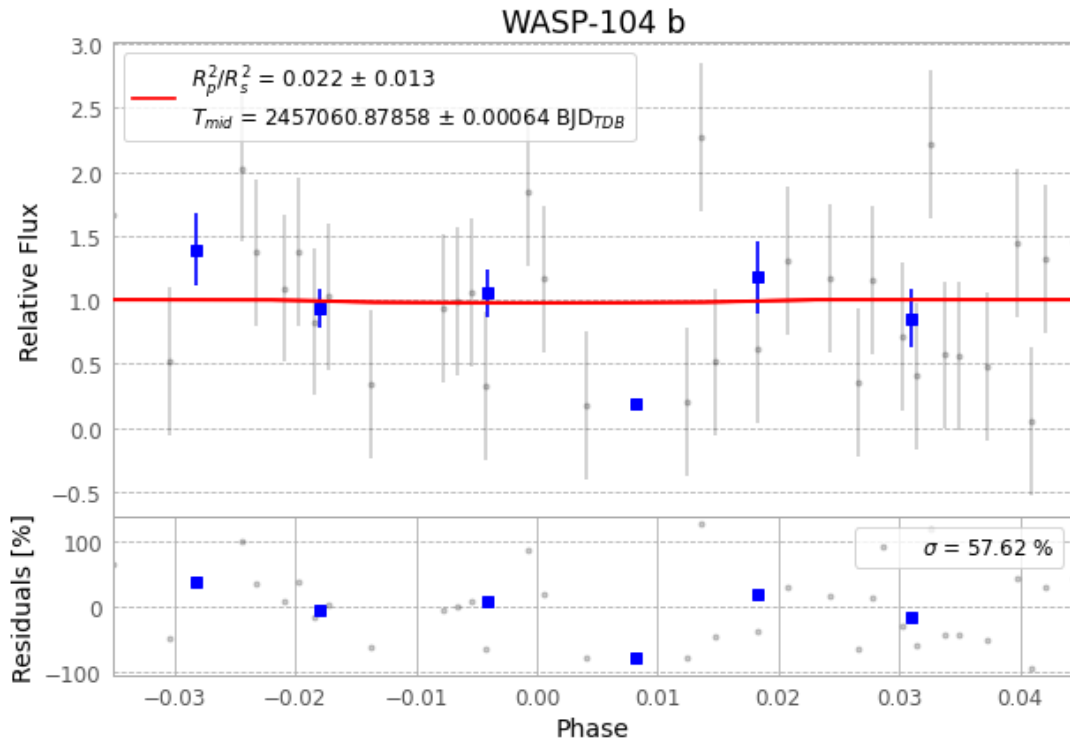


Figure 1 — FinalLightCurve_WASP-104 b_2015-02-07.png (SHA-256 a8b5b387eddc393e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [54, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b3050f2154b6013d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), WASP-104150207072713.FITS → WASP-104150207110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150207.log (731b77a4945d...), FinalLightCurve_WASP-104 b_2015-02-07.png (a8b5b387eddc...), FinalLightCurve_WASP-104 b_2015-02-07.csv (625185d17730...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:45Z.